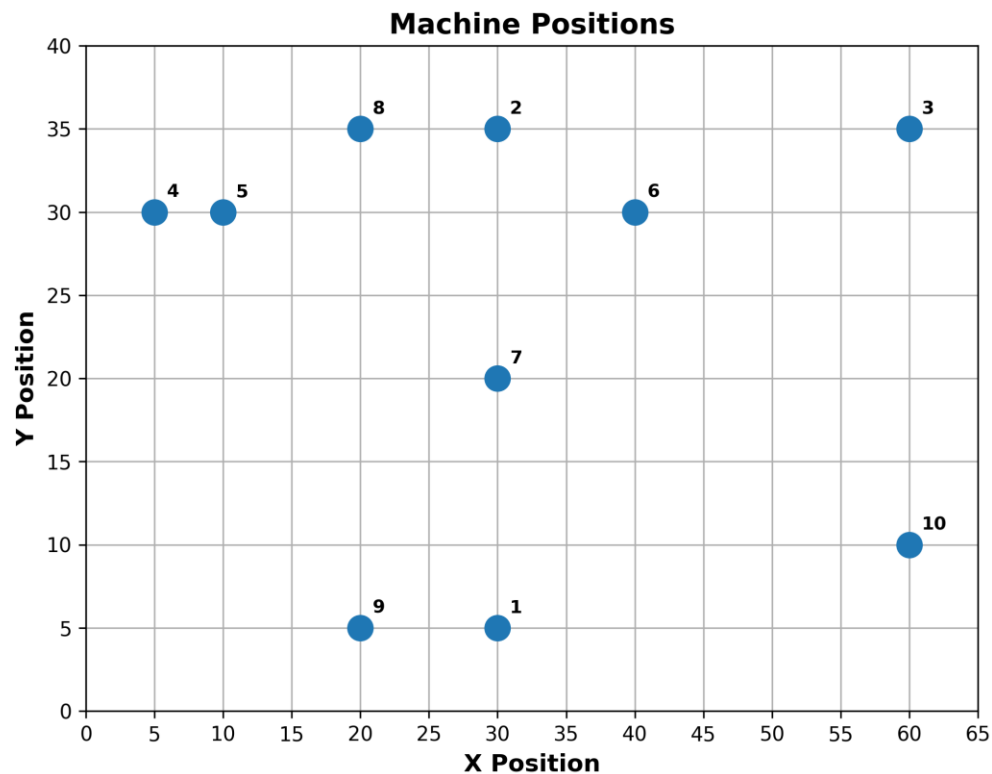
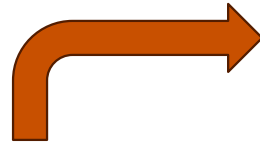


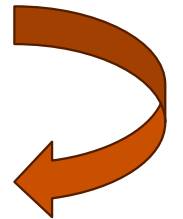
Anh Le Xuan, Moritz Engelmann, Max Berthold, Henrik Stein

Logistics Lab WiSe 2025/26

Aufgabe 1

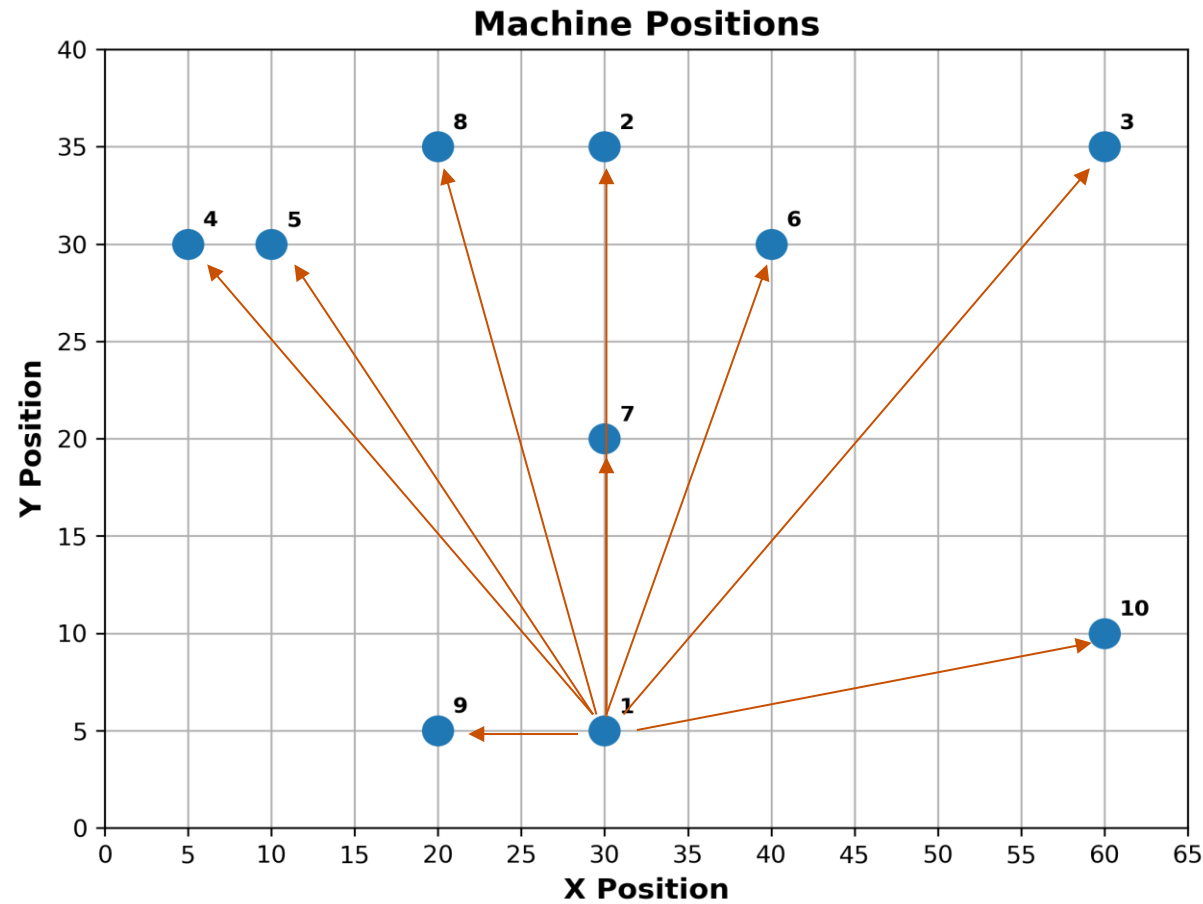


Transport Demand		
start	dest	number
1	2	7
1	3	1
1	4	1
1	5	9
1	6	1
1	7	8
1	8	9
1	9	2
1	10	7
2	1	5

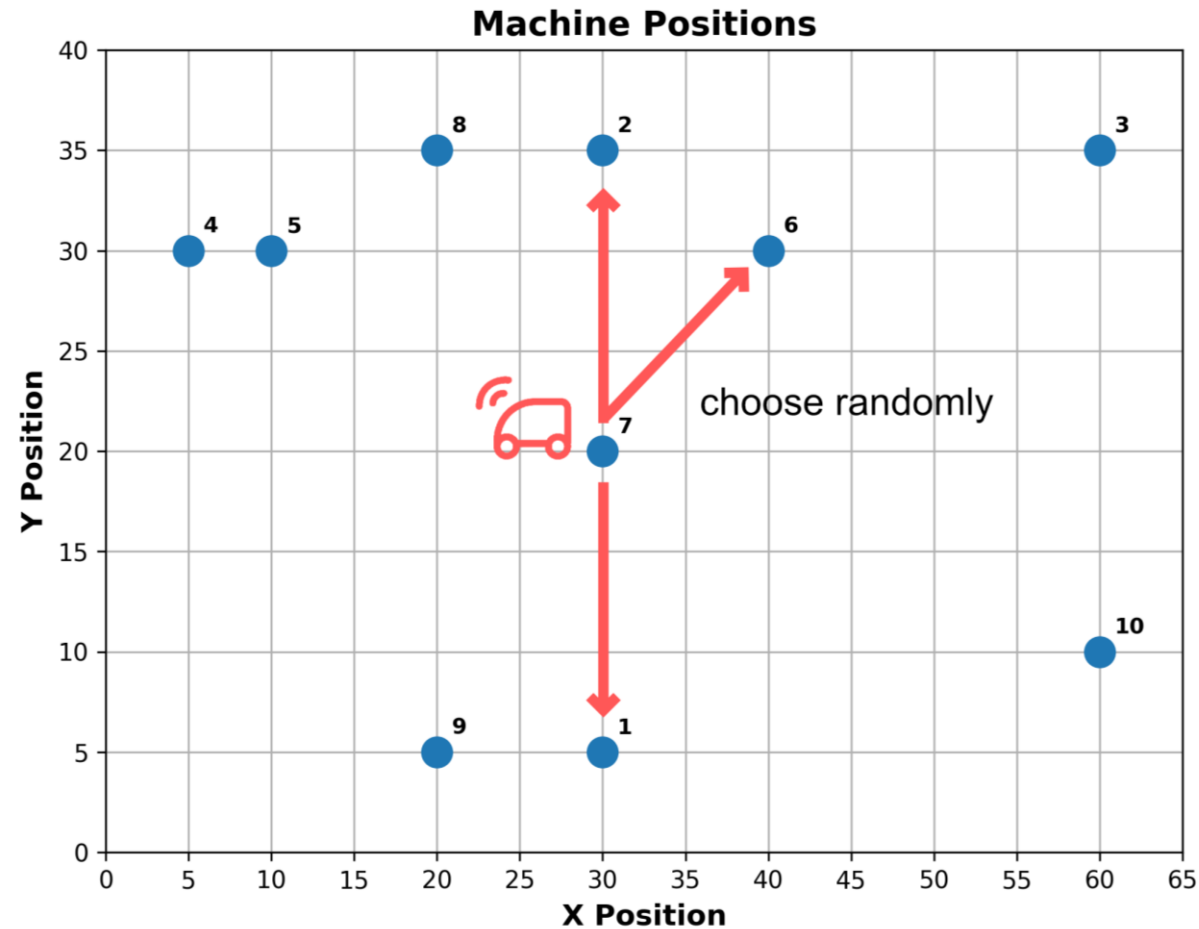


Schedule Example				
VehicleId	Location	unload	load	emptyRuns
1	10	0	1	77
1	4	1	0	77
1	10	0	1	77
1	4	1	0	77
2	2	0	1	77
2	8	1	1	77
2	2	1	1	77
2	8	1	1	77
2	2	1	1	77
2	8	1	1	77

Bruteforce



k-Nearest-Neighbor (kNN)



Vehicle Routing Problem

2. Mathematical Formulation the VRP

Let x_{ij} be an integer variable which may take value $\{0, 1\}$, $\forall \{i, j\} \in E \setminus \{0, j\} : j \in V$ and value $\{0, 1, 2\}$ $\forall \{0, j\} \in E, j \in V$. Note that $x_{0j} = 2$ when a route including the single customer j is selected in solution

VRP can be formulated as follows :

$$\min \sum_{i \neq j} d_{ij} x_{ij}$$

Subject to:

$$\sum_j x_{ij} = 1, \forall i \in V,$$

$$\sum_j x_{ij} = 1, \forall i \in V,$$

$$\sum_i x_{ij} |S| - c(S), \{S : S \subseteq V \setminus \{1\}, |S| \geq 2\},$$

$$x_{ij} \in \{0, 1\}, \forall \{i, j\} \in E; i \neq j$$

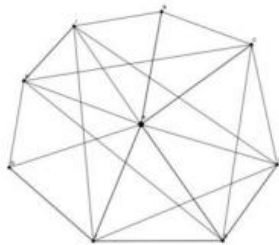


Figure 1. VRP case

Figure 1 shows an example of a VRP case. There are 8 points that will be visited by 2 vehicles to deliver goods with a certain distance between points. Point A as a depot, and then point B,C,D,E,F,G,H,I as the location to be visited

Paper: Study vehicle routing problem using Nearest Neighbor Algorithm

Autor: Rio Ferdiani Harahap and Sawaluddin

Jahr: 2023

Quelle: Journal of Physics: Conference Series

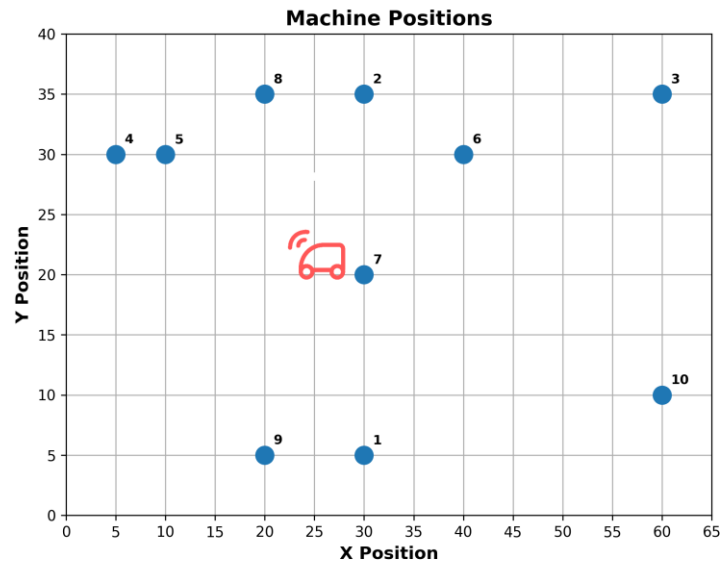
Konferenz: 2nd TALENTA-International Conference on Science and Technology

https://www.researchgate.net/publication/367415289_Study_vehicle_routing_problem_using_Nearest_Neighbor_Algorithm

kNN - Variante 1

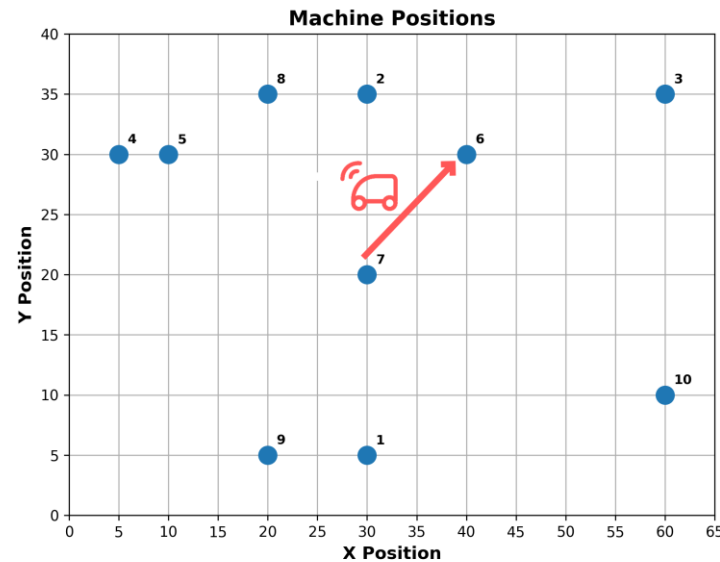
Greedy Heuristic (Algorithm)

GH



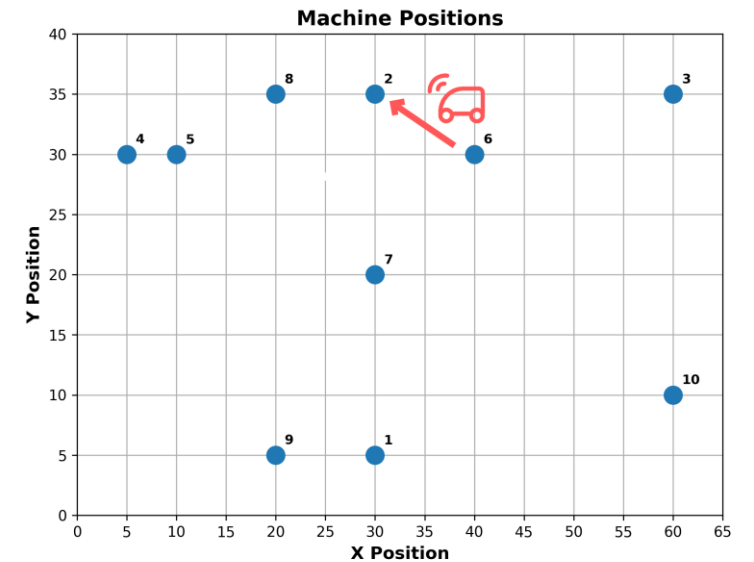
Greedy Heuristic (Algorithm)

GH



Greedy Heuristic (Algorithm)

GH

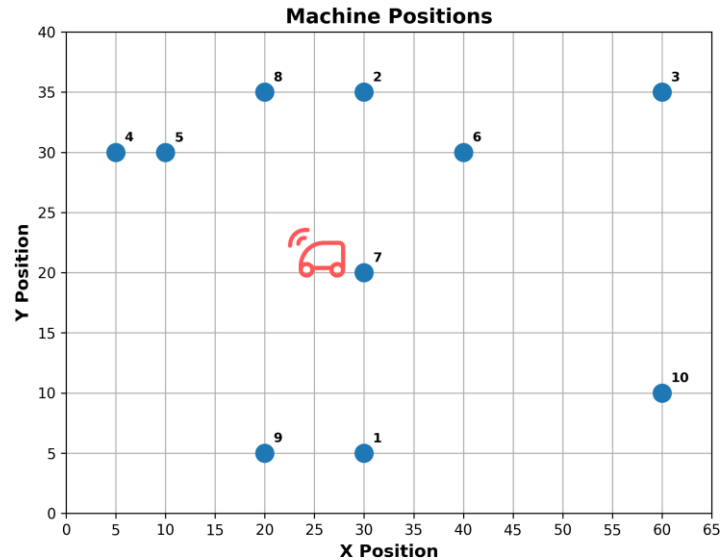


kNN - Variante 2

Randomized Greedy Heuristic

RGH

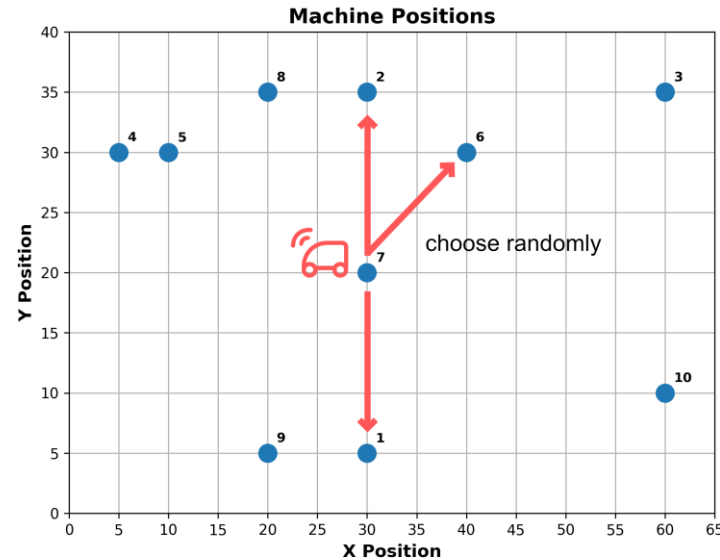
e.g. $k = 3$



Randomized Greedy Heuristic

RGH

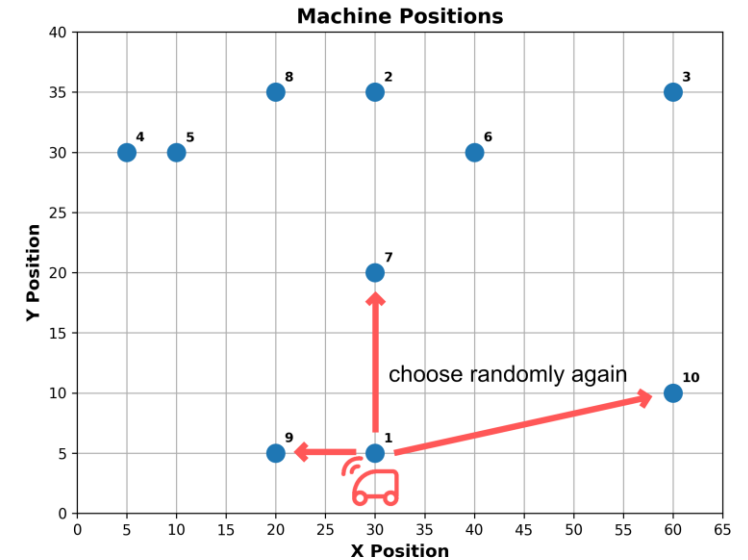
e.g. $k = 3$



Randomized Greedy Heuristic

RGH

e.g. $k = 3$

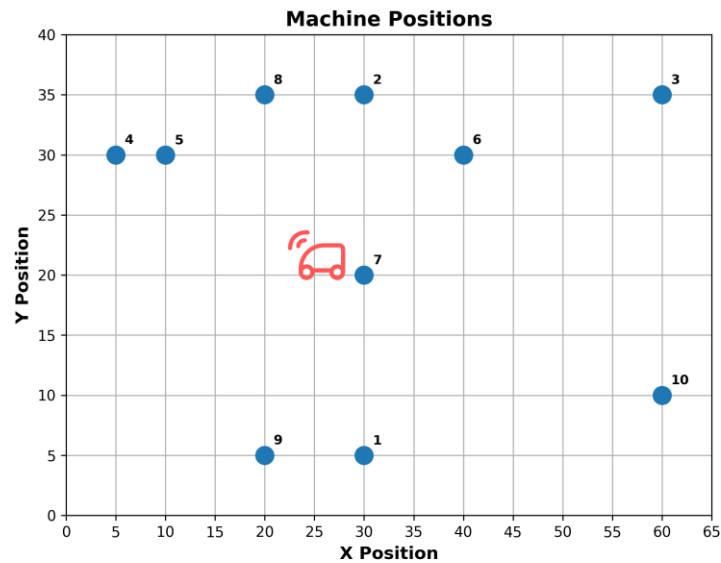


kNN – Variante 3

Multistart Randomized Greedy Heuristic

MRGH

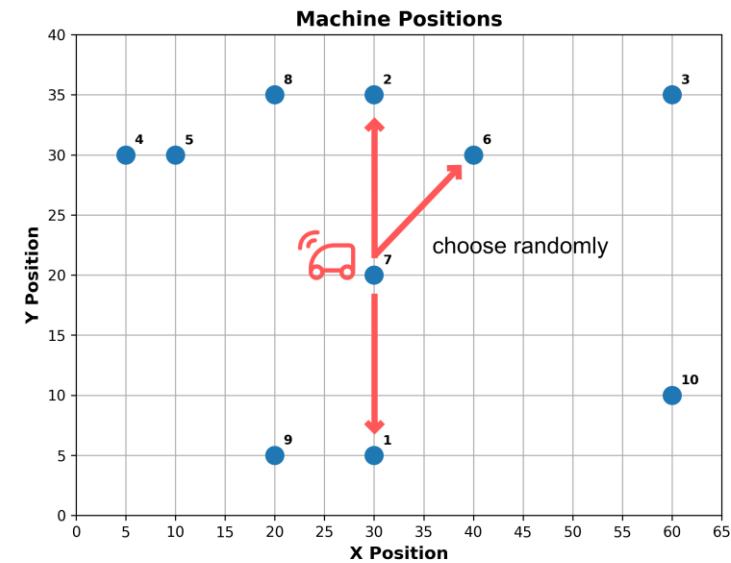
e.g. $k = 3$



Multistart Randomized Greedy Heuristic

MRGH

e.g. $k = 3$



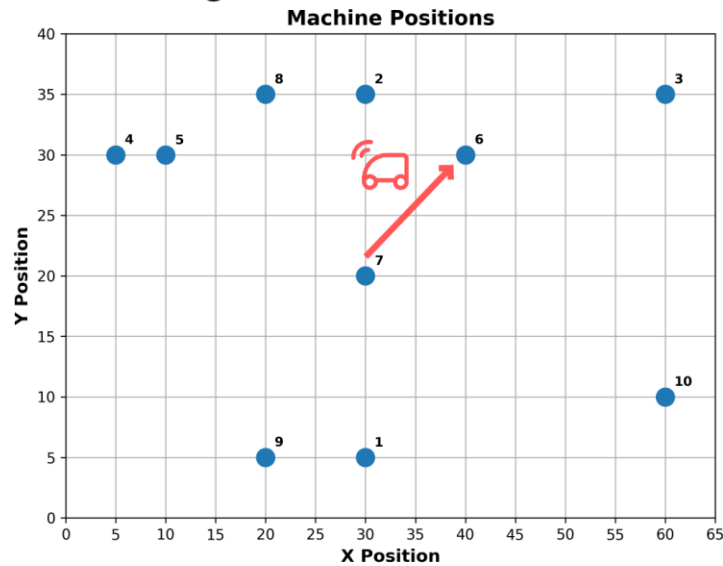
kNN – Variante 3

Multistart Randomized Greedy Heuristic

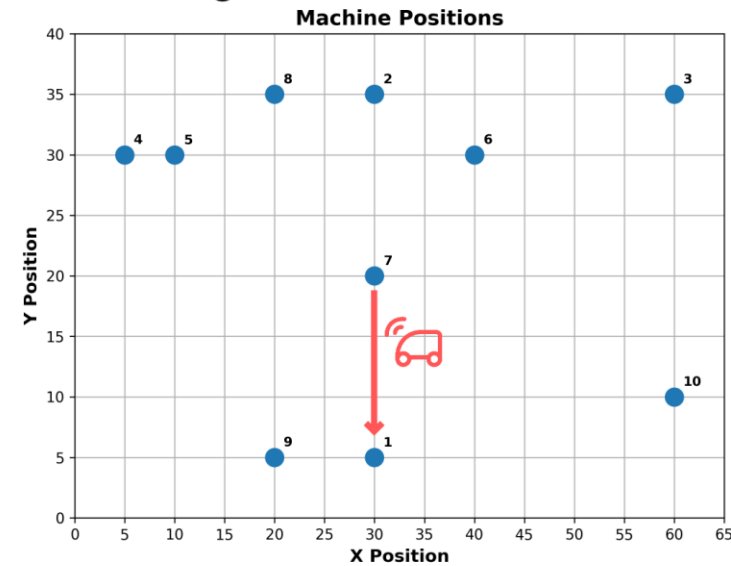
MRGH

e.g. $k = 3$

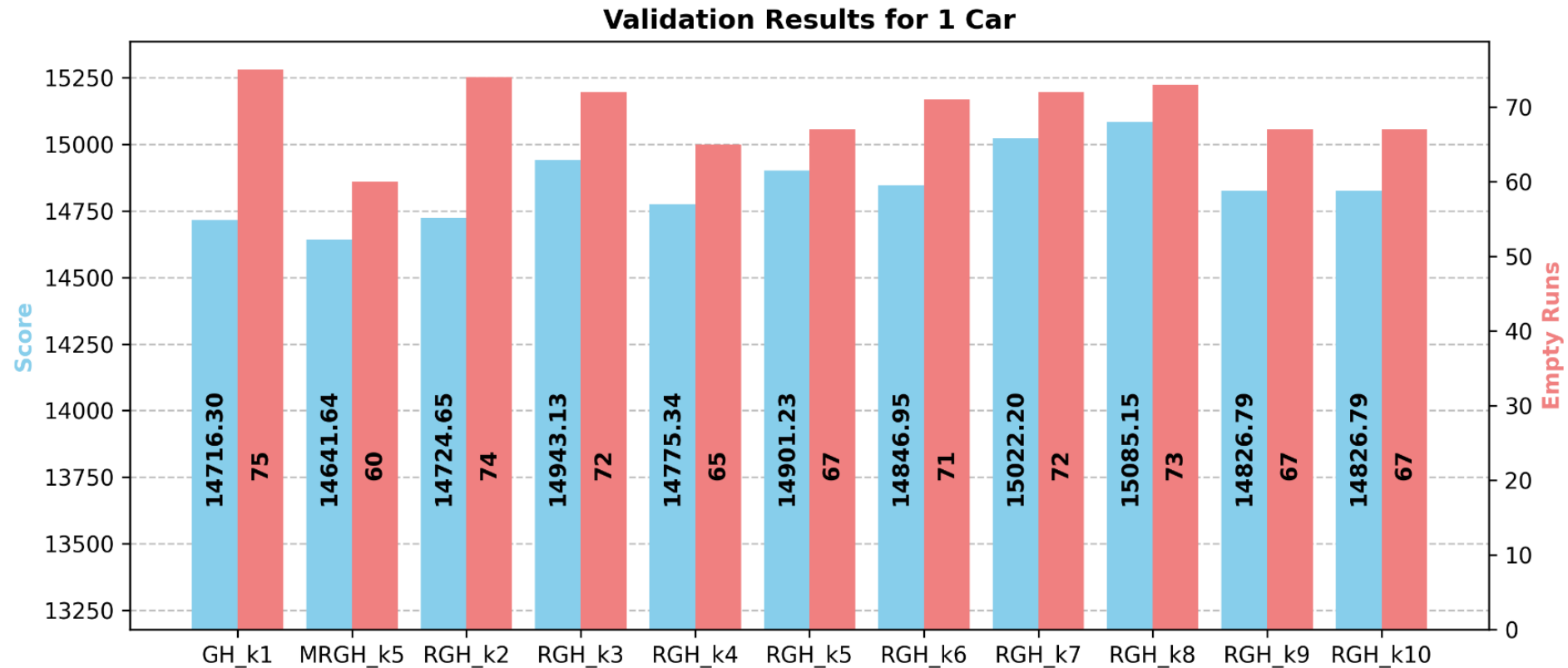
e.g. random seed = 1



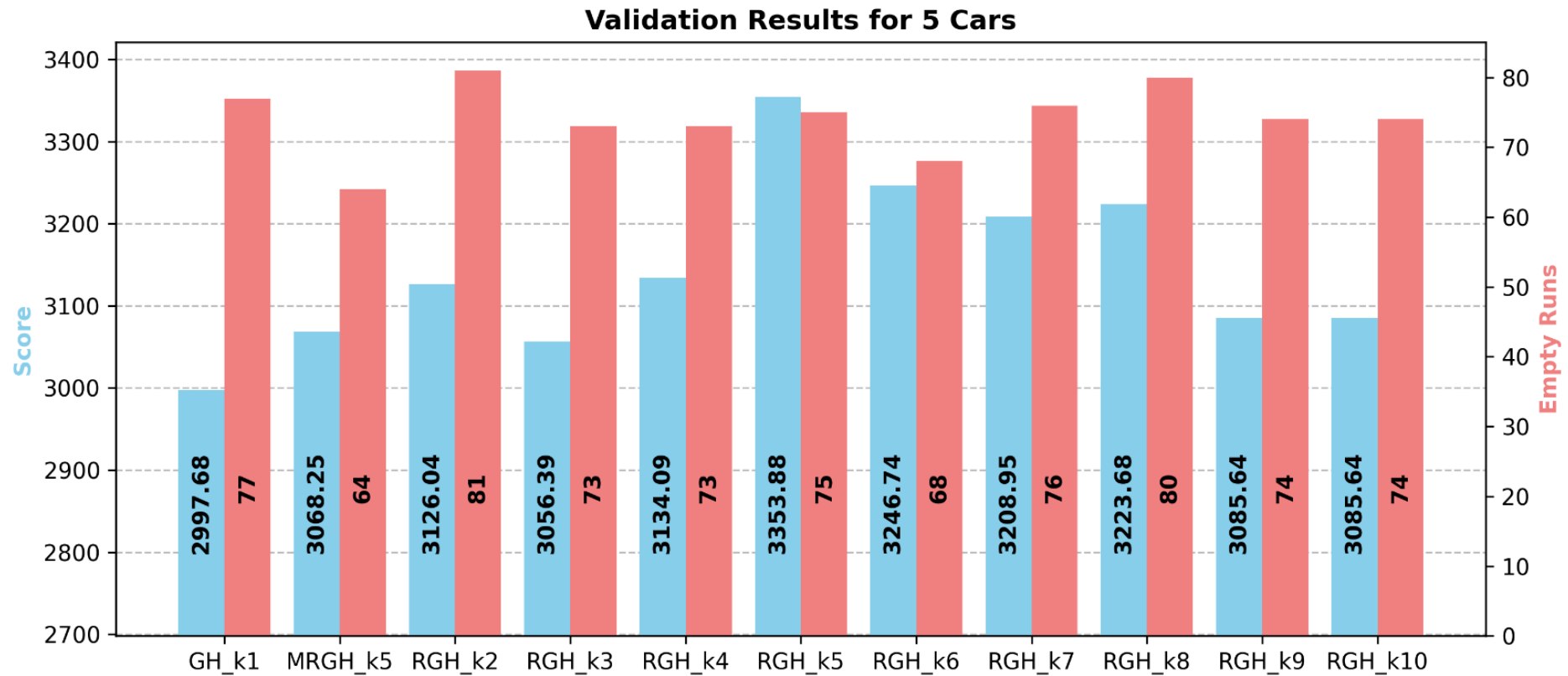
e.g. random seed = 2



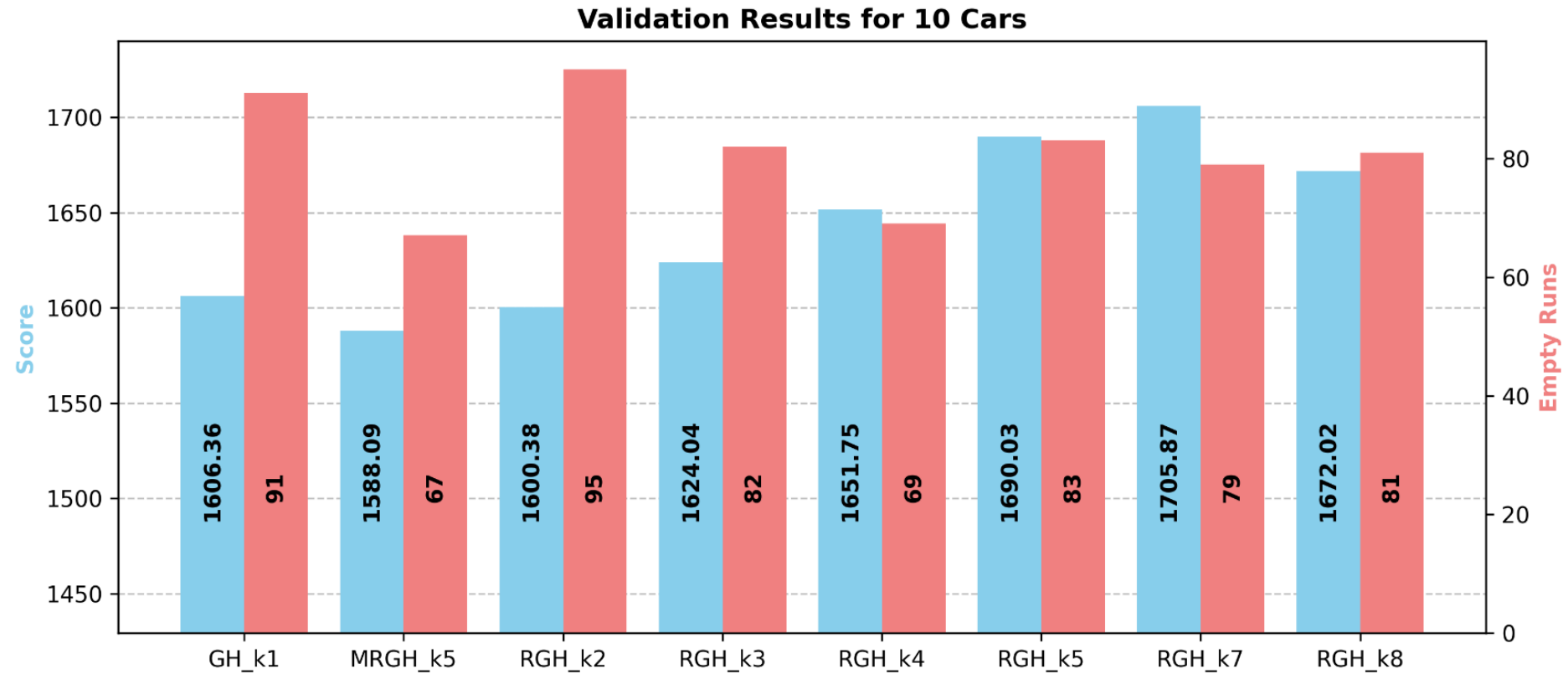
kNN - Ergebnisse



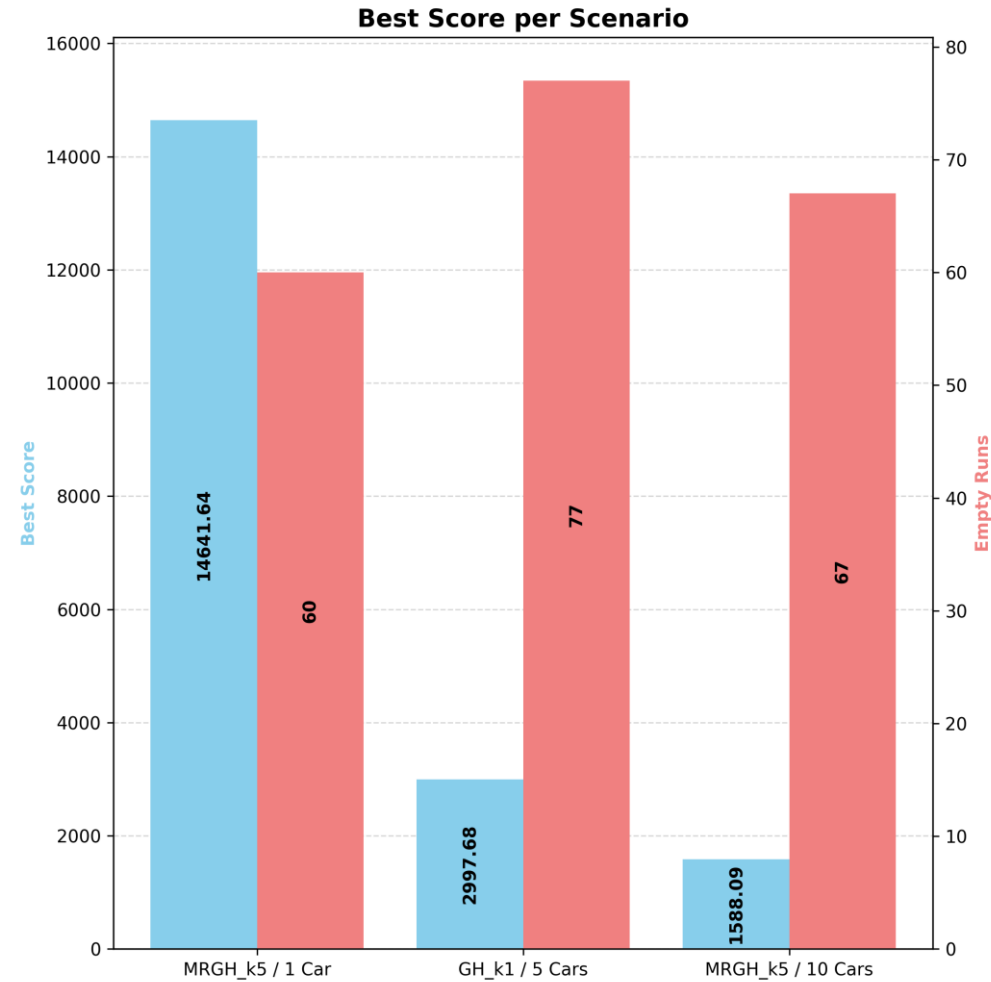
kNN - Ergebnisse



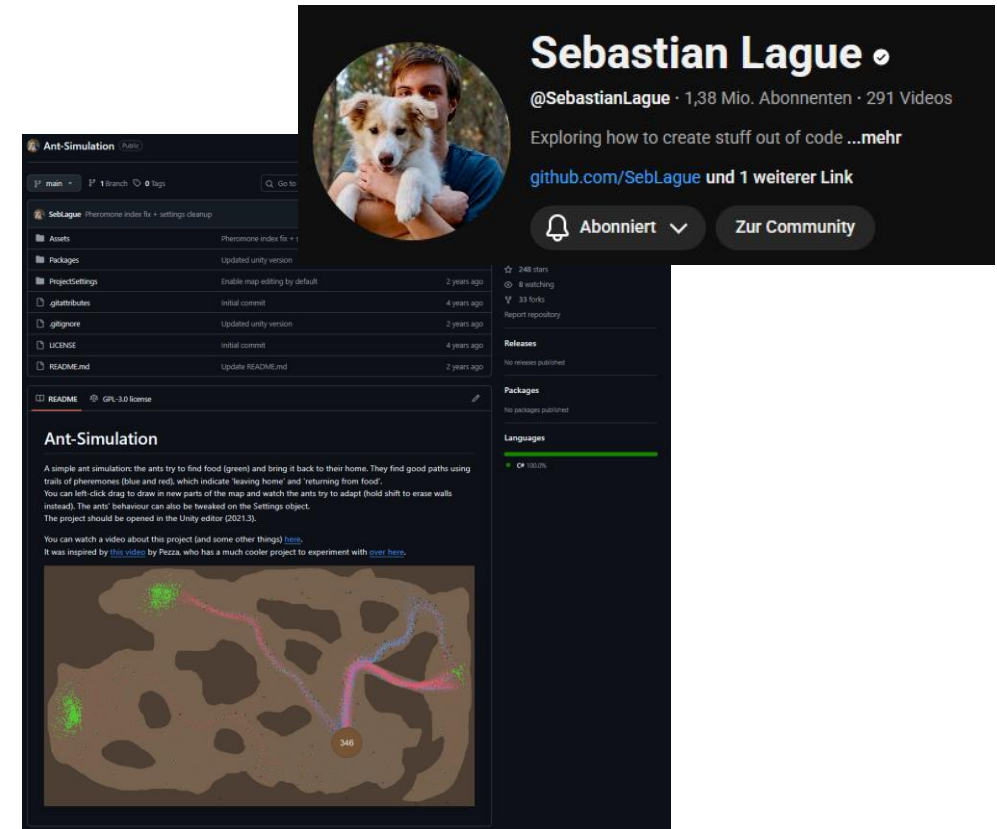
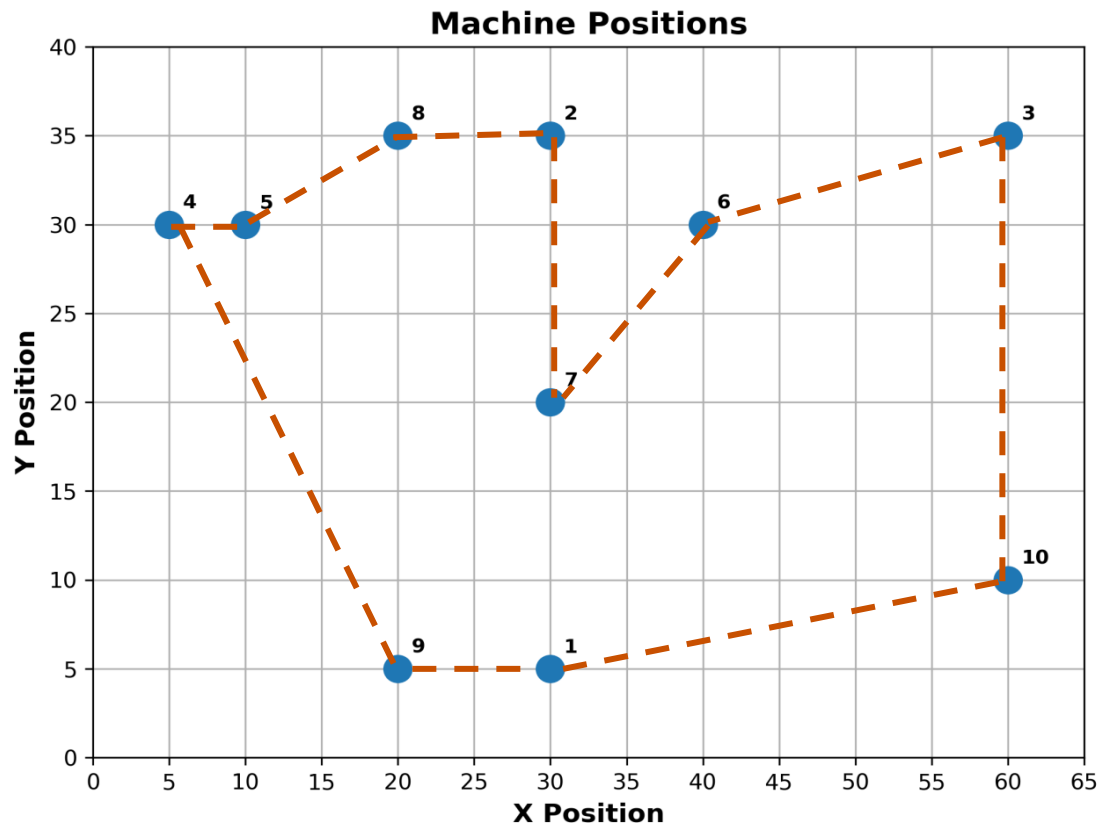
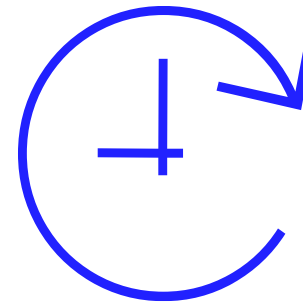
kNN - Ergebnisse



kNN – Vergleich

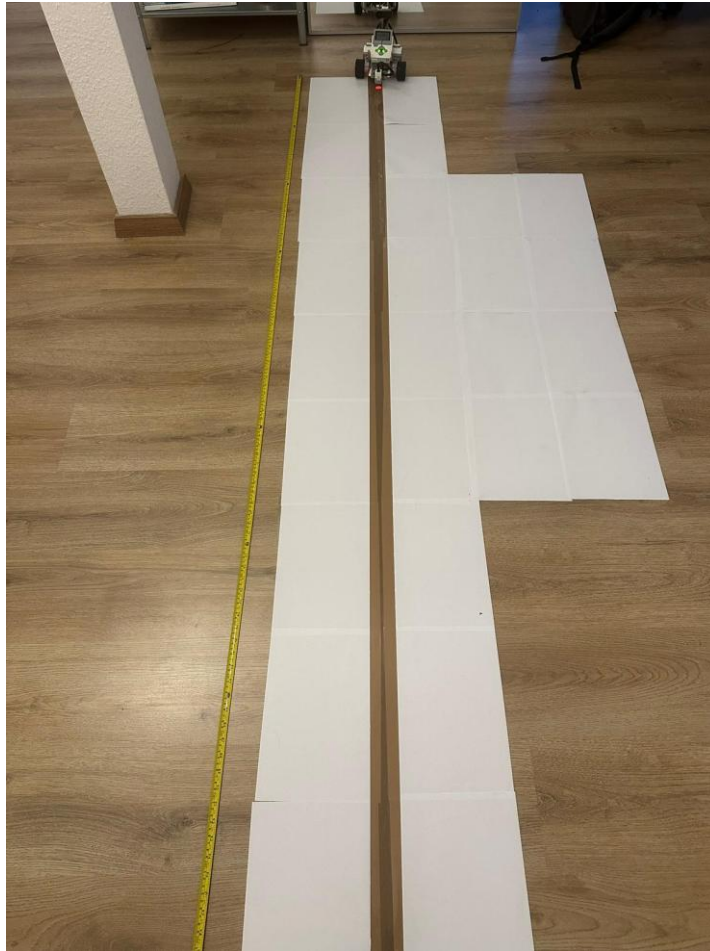


Ant Colony Algorithm



<https://github.com/SebLague/Ant-Simulation>

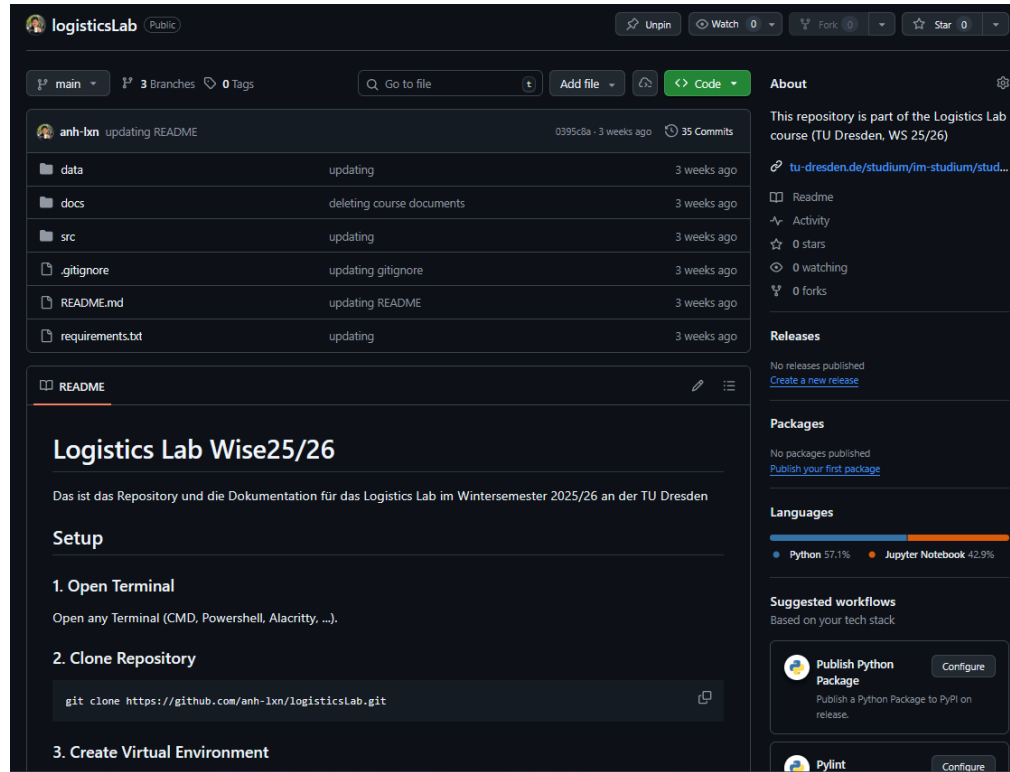
Aufgabe 2



Umfahren eines Hindernisses

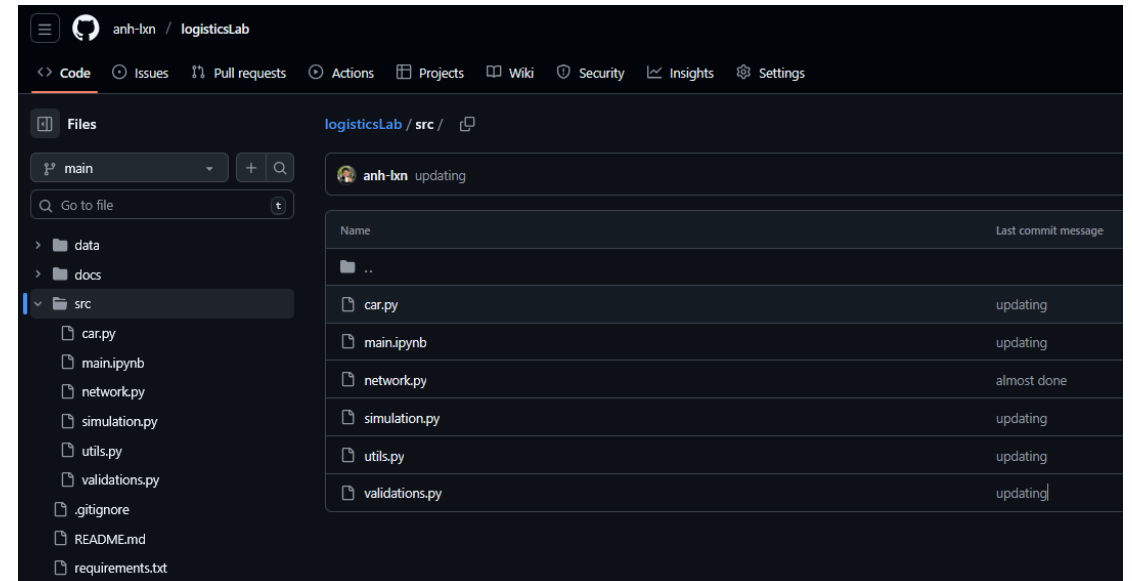


Dokumentation



<https://github.com/anh-lxn/logisticsLab>

Code



<https://github.com/anh-lxn/logisticsLab/tree/main/src>

Literatur & Quellen

❖ **Harahap, R. F., & Sawaluddin (2023):**

Study vehicle routing problem using Nearest Neighbor Algorithm.

Journal of Physics: Conference Series, 2421, 012027.

doi: 10.1088/1742-6596/2421/1/012027

ReserachGate: https://www.researchgate.net/publication/367415289_Study_vehicle_routing_problem_using_Nearest_Neighbor_Algorithm

❖ **Lague, S. (2021):**

Coding Adventure: Ant and Slime Simulations

YouTube: Sebastian Lague: <https://www.youtube.com/@SebastianLague>

❖ **Weiner, A. (2021)**

ML-CFD Lecture – Surrogate Modeling for Discrete and Continuous Predictions (Lectures 4&5)

GitHub-Repository: <https://github.com/AndreWeiner/ml-cfd-lecture>

❖ **OpenAI (ChatGPT) & GitHub Copilot (2025)**

Einsatz als Hilfsmittel für Code-Nachfragen, Code-Cleanup und Erläuterungen; keine inhaltliche Autorenschaft